MRI-ReportGenerator: An Application Pipeline for Automated Cervical Spine MRI Measurement and Structured Reporting

A segmentation-to-report system with literature-grounded thresholds and a threshold-crossing validation against healthy and pathological cohorts

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Abstract

Cervical spine MRI is read manually by radiologists, a process that is slow and subject to high inter-observer variability in the quantitative measurements (vertebral heights, disc dimensions, canal and cord diameters, sagittal alignment) that drive clinical decisions in degenerative cervical myelopathy and radiculopathy. We present **MRI-ReportGenerator**, an application-positioned pipeline that converts a single sagittal T2 cervical MRI (DICOM or NIfTI) into a structured, radiology-style report. The system chains three pre-trained, openly licensed segmentation engines (TotalSpineSeg, the Spinal Cord Toolbox, and SPINEPS), a suite of geometric and signal-based measurement modules organized into five groups (vertebral body, intervertebral disc, canal/cord, sagittal alignment, and signal/shape screens), and an interpretation layer that classifies every measurement against a central catalog of literature-grounded thresholds and emits findings "flagged for physician review," never a diagnosis. A core engineering principle is that every measurement is a disease-agnostic geometric or signal detector anchored on healthy normative values, so that the abnormal-detection generalizes across any pathology that produces the same anatomical change, rather than being trained on a particular disease. Because no public dataset pairs cervical MRI with per-case expert measurements, we validate by *threshold crossing and distribution separation*: healthy necks must sit on the normal side of each cited threshold, and a symptomatic cohort must shift into the abnormal side. On 12 healthy Spine-Generic controls versus a symptomatic cervical-spondylosis cohort from the MMCSD dataset (scaled to 41–49 cases for the disc and alignment groups), the canal/cord measurements separate strongly (functional canal AP minimum and space-available-for-cord, both Mann–Whitney $p = 0.0001$) and the vertebral compression screen is correctly silent on a non-compression cohort. Two leads were *revised* by larger, design-appropriate cohorts, which we report as a methodological result: the SPINEPS endplate-voxel Cobb method reads correct lordosis (15.2°) but on a representative multi-site healthy sample does *not* discriminate myelopathy (a validated measurement, not a screen), and the disc group — after the harness detected and we fixed a sign-level height/bulge bug — separates only modestly, through disc *geometric spread* (disc/vertebral-body AP ratio, AUC 0.62), with disc signal and bulge documented as non-discriminators. Four measurement defects were found and fixed, and the interpretation layer was wired and run end-to-end with patient-demographic-aware thresholds. We document the full mistake-to-fix methodology narrative, the limitations imposed by the absence of ground truth, and the deployment architecture. The contribution is less a new algorithm than a disciplined, honest, end-to-end *engineering* of clinical measurement automation with explicit provenance and validation.

Contents

1 Introduction

1.1 Motivation

Degenerative cervical spine disease is among the most common reasons a cervical MRI is acquired. Degenerative cervical myelopathy (DCM) is the leading cause of spinal-cord dysfunction in adults worldwide, and cervical spondylotic radiculopathy is a frequent cause of arm pain and disability. The radiologist's report on a cervical MRI is built on a set of quantitative measurements: the heights and shape of each vertebral body, the dimensions and signal of each intervertebral disc, the antero-posterior diameter of the spinal canal and the cord and the space available between them, the sagittal alignment (lordosis), and the presence of abnormal cord signal. These numbers are then compared, often implicitly, against normative values to decide whether a finding is present and how severe it is.

Performed manually, this is slow and variable. Measuring a single disc height, a canal diameter, or a Cobb angle by hand requires careful caliper placement on the right slice, and studies of inter-observer agreement repeatedly show that different readers — and the same reader on different days — produce materially different numbers. The non-AI baseline our application competes against is therefore a manual workflow that is both labour-intensive and noisy.

1.2 What this system does

MRI-ReportGenerator takes one sagittal T2 cervical MRI and produces a structured report. The pipeline is:

`Input → Segmentation → Measurements → Interpretation → Report`

Segmentation is delegated to three established, openly licensed deep-learning tools, each chosen for the anatomical structure it segments best. The measurement layer is organized into five groups that mirror the structures a radiologist reports on. The interpretation layer compares each measurement to a central catalog of thresholds, each carrying an explicit literature citation and a modality caveat. Every output is phrased as a finding "flagged for physician review"; the system never asserts a diagnosis.

1.3 Positioning: an application, and an honest one

This project is positioned as an *application*, not a research contribution claiming a new state-of-the-art algorithm. Its value is in the disciplined engineering: composing validated tools, grounding every clinical number in a primary source, and — crucially — validating the assembled pipeline against real healthy and pathological data with a methodology appropriate to the available ground truth. Two principles run through the work:

1. **Measurements are disease-agnostic geometry/signal detectors anchored on healthy norms.** A canal narrowed to 8 mm reads as stenosis whether the cause is spondylosis, tumour, or trauma. The normal range for each measurement is taken from healthy-population literature, not learned from a disease cohort. Consequently the system's abnormal-detection generalizes across pathologies that produce the same anatomical change, and we never overfit to a particular disease.

2. **Honesty about uncertainty.** Where a threshold is borrowed from a different modality (e.g. a radiograph-derived ratio applied to MRI), or where no cited value exists, the system says so explicitly rather than inventing a number.

1.4 The validation problem

The central difficulty is that *no public dataset pairs cervical MRI images with per-case expert measurements*. We confirmed this across repeated, exhaustive dataset hunts. Without such ground truth we cannot compute a true sensitivity or specificity against a radiologist. We therefore adopt a **threshold-crossing and distribution-separation** design: on healthy necks every measurement must sit on the normal side of its cited threshold and never cross; on a symptomatic cohort the distribution must shift into the abnormal side, with visibly pathological cases flagging. This demonstrates that each measurement "reads the right range" without claiming per-case diagnostic accuracy.

1.5 Contributions

- An end-to-end, deployable cervical-MRI measurement and reporting pipeline composing three segmentation engines, five measurement groups, and a cited-threshold interpretation layer.
- A central, fully cited threshold catalog with explicit modality caveats and a no-diagnosis output ontology.
- A literature-grounded methodology for the two hardest measurements (the vertebral endplate/corner geometry and the cervical Cobb angle), including the discovery that fitting a line to SPINEPS' own predicted endplate voxels outperforms both corner-extrema and corpus-border methods.
- A reproducible threshold-crossing validation on real cervical MRIs with non-parametric statistics, showing strong separation for the canal/cord group, a correctly null result for the vertebral compression screen on a non-compression cohort, the detection of a real sign-level bug in the disc-height index, and — on scaled, balanced cohorts — the honest revision of an initial alignment lead to a *non-discriminator* (and the disc signal/bulge axes to negatives).
- A transparent "development journey" documenting every substantive mistake, how it was found, the fix, and how the fix was validated — the methodological core of the project.

1.6 Paper organization

Section 2 reviews the clinical background and the measurements. Section 3 surveys the segmentation tools, prior measurement methods, and datasets. Section 4 describes the system architecture. Section 5 details the datasets. Sections 6–12 present the segmentation and the five measurement groups plus the interpretation layer. Section 13 formalizes the validation methodology and Section 14 reports the results. Section 15 narrates the development journey, Section 16 the limitations, Section 17 the deployment, and Section 18 concludes.

2 Clinical Background

2.1 Cervical anatomy on sagittal T2 MRI

The cervical spine comprises seven vertebrae (C1–C7) and the intervening discs (C2–C3 through C7–T1). On a mid-sagittal T2 image, the bright cerebrospinal fluid (CSF) outlines the thecal sac, with the spinal cord visible as an intermediate-signal band within it. We measure C3–C7 and the discs between them; C1 (atlas) and C2 (axis/odontoid) are excluded because their morphology is structurally unique and the standard vertebral-body measurements do not apply.

2.2 The measurements and what they mean clinically

Vertebral body (Group 1). Anterior, middle, and posterior heights (H_a, H_m, H_p); the compression ratio $\mathrm{H_a/H_p}$; antero-posterior (AP) width; tilt; and spondylolisthesis (slip). A reduced $\mathrm{H_a/H_p}$ indicates anterior wedging (compression/deformity); a Genant-style grade describes the degree [6]. Spondylolisthesis is forward/backward slip of one vertebra on another.

Intervertebral disc (Group 2). Disc signal-intensity height, the disc-height index (DHI), posterior disc bulge, and a signal-degeneration grade. Disc-height loss and signal loss are hallmarks of degeneration; a posterior bulge or herniation can compress the cord or nerve roots. Cervical disc signal is graded by the modified (Miyazaki) scheme [9], not the lumbar Pfirrmann scheme.

Canal and cord (Group 3). The functional (soft-tissue) canal / dural-sac AP minimum, the spinal cord AP diameter, the space available for the cord (SAC = canal − cord), and the Torg–Pavlov ratio. Canal narrowing and a small SAC are the structural substrate of myelopathy; the cord may flatten under compression. Normative cervical canal, SAC, and vertebral-width values are available from population studies [1], and cord cross-section can be compared to a healthy normative database [16].

Sagittal alignment (Group 4). The C2–C7 (and C3–C7) Cobb angle, segmental angles, and a lordosis classification. The cervical spine is normally lordotic; loss of lordosis or kyphosis is associated with degeneration and with myelopathy outcomes. Asymptomatic C2–C7 lordosis is roughly 10° to 35° [2].

Signal and shape screens (Group 5). A vertebral-body compression/deformity screen and a cord T2-hyperintensity (myelomalacia) screen. Intramedullary T2 hyperintensity is a marker of myelopathy.

2.3 Two clinical syndromes the cohort represents

Cervical spondylosis (degenerative “wear-and-tear”) produces two principal syndromes that our unhealthy cohort is labelled by:

- **Cervical Spondylotic Myelopathy (CSM)** — spinal-cord dysfunction from canal stenosis and cord compression. Structurally this is a Group 3 (canal/SAC/cord) and Group 5.1 (cord signal) disease.
- **Cervical Spondylotic Radiculopathy (CSR)** — nerve-root compression, usually from disc herniation/osteophyte. Structurally this is a Group 2 (disc) disease.

Crucially, *spondylosis is not the same as vertebral compression fracture*. Compression fractures arise from osteoporosis or trauma, not degeneration, so a spondylosis cohort is expected to have normal vertebral-body heights. This distinction (spondyl*osis* vs. spondyl*olisthesis* vs. compression) is central to interpreting our validation: it explains why the Group 1 compression screen should be, and is, silent on this cohort (Section 14).

2.4 The no-diagnosis constraint

Quantitative measurements are necessary but not sufficient for diagnosis: an abnormal value can occur in asymptomatic individuals (for example, a large fraction of asymptomatic adults have a measurable disc bulge or a low Torg ratio), and clinical correlation is always required. The system therefore reports measurements and threshold crossings as findings “flagged for physician review," never as diagnoses — a hard rule enforced throughout the codebase and the output wording.

3 Related Work

3.1 Spine segmentation tools

Three openly licensed, pre-trained engines underpin our pipeline. **TotalSpineSeg** [18] (LGPLv3) is an nnU-Net–based tool that labels individual vertebrae and discs (and a canal/cord region) on spine MRI, producing per-structure integer labels in a canonical space. **The Spinal Cord Toolbox (SCT)** [5] (LGPLv3) provides validated deep-learning segmentation of the spinal cord and the dural-sac/canal (`sct_deepseg`), per-slice morphometry (`sct_process_segmentation`), a normative healthy-control cord database (PAM50, via `-normalize-hc`) [16], and the SCIseg model for intramedullary lesion segmentation [12]. **SPINEPS** [10] segments vertebrae with explicit endplate prediction; as we show, its endplate voxels are the key to an accurate Cobb angle. We use each tool for the structure it segments best rather than treating them as interchangeable.

3.2 Vertebral endplate/corner geometry and the Cobb angle

Vertebral-body heights and the Cobb angle both depend on locating the vertebral endplates. The classic approach extracts four (or six) corner landmarks and measures between them, but corner-extrema methods are unstable on real lordotic cervical necks because cervical endplates are concave and sloped: the most-posterior voxel in a flat band lands on the concave interior, mislocating the corner [3]. The one head-to-head cervical study, Wang et al. [17], shows that *fitting a line to the full endplate boundary* beats four-corner extrema (ICC 0.97 vs. 0.75; MAE 3.23° vs. 5.42°). The accuracy ceiling for an automated cervical-T2 Cobb is set by Zhang et al.'s endplate-line nnU-Net (ICC 0.94, MAE 2.44°) [20]. These findings directly drive our methods: we replace corner extrema with robust endplate-line fitting, and ultimately fit the line to SPINEPS' own predicted endplate voxels.

3.3 Normative values and thresholds

Our thresholds are grounded in primary sources: healthy cervical vertebral H_a/H_p and body heights [15, 8, 7]; healthy cervical canal, SAC, Torg, and vertebral-width percentiles [1]; cord cross-section normatives [16]; disc bulge thresholds [13]; the cervical (Miyazaki) disc-degeneration grade [9]; asymptomatic cervical lordosis [2]; and the Genant vertebral-deformity grade as a thoracolumbar-derived standard used with explicit caveats [6]. Where a commonly cited threshold is in fact a radiograph- or CT-derived value (Torg–Pavlov; some canal cut-points), we flag it as such because it can over-flag on MRI soft-tissue measurements.

3.4 Datasets

Public spine datasets with usable cervical MRI are scarce. **Spine-Generic** [4] provides multi-vendor healthy-control cervical T2 acquisitions and is our healthy anchor. **MMCSD** [19] (Yu et al., *Scientific Data*) provides 250 surgical cervical-spondylosis cases with real NIfTI volumes and per-case CSM/CSR labels plus per-level lesion flags — the labelled symptomatic cohort we use. **Duke CSpineSeg** [21] provides clinical cervical T2 with expert vertebra/disc masks but no per-case pathology labels and no per-case measurements. We confirmed across repeated searches that no public dataset pairs cervical MRI with per-case expert *measurements*; this absence shapes our validation design.

4 System Architecture

4.1 Pipeline overview

The system is structured as an external entry-point (EEP) orchestrating internal processing services (IEPs), each a containerized stage. Figure 1 shows the flow.

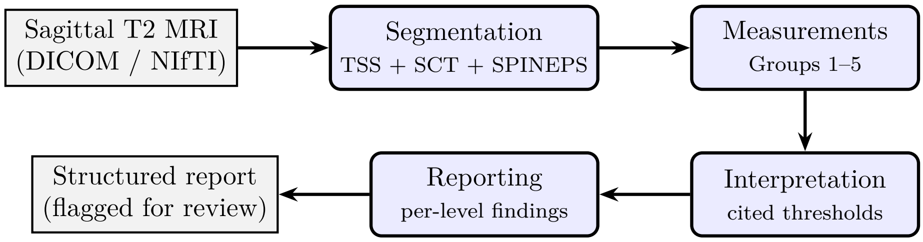


Figure 1: End-to-end pipeline. Segmentation delegates to three engines; measurements are organized into five groups; interpretation classifies each measurement against a cited-threshold catalog; reporting emits per-level findings flagged for physician review.

4.2 Services

- `services/segmentation/` — TotalSpineSeg wrapper (vertebra/disc, `-iso`) and a separated Spinal Cord Toolbox path (canal/cord). SPINEPS runs in its own environment due to a numpy pin (`numpy==2.0.2`) that conflicts with the TSS stack.
- `services/measurements/` — the measurement runtime. `geometric/` holds Groups 1, 2, and 4 (vertebral morphometry, disc, alignment) plus vendored validated geometry (`_vertebral_geometry.py`, `_endplate_cobb.py`); `cord/` holds Group 3 (canal/cord/SAC via SCT); `signal/` holds disc signal grading; `group5/` holds the signal/shape screens; `orchestrator.py` runs the registered components in dependency order with Prometheus instrumentation.
- `services/interpretation/` — the cited-threshold catalog (`thresholds.py`) and the interpretation engine (`interpretation.py`).
- `services/reporting/` — report assembly and rendering; `services/eep/` — the public API entry point.

4.3 The MeasurementContext and canonical orientation

A per-case `MeasurementContext` loads the segmentation once, reorients it to canonical RAS (so downstream code can rely on axis 0 = left→right, axis 1 = posterior→anterior, axis 2 = inferior→superior), enforces a 1 mm isotropic measurement grid, and records a resolution-quality flag that warns when the acquisition slice thickness was coarse (typical cervical T2 is 3 mm to 4 mm through-plane, so the 1 mm grid is interpolated, not real detail). This canonical context is the input contract for every measurement component.

4.4 Output data contract

Each component returns a standard `ComponentResult` (`measurements`, `flags`, `intermediate`, `metadata`). The orchestrator flattens these into `{components, measurements, flags, interpretations}`, and the interpretation layer wraps each (measurement × level) into a container carrying `status` (`within_reference` / `outside_reference` / `review_only` / `not_interpretable`), a per-measurement `severity`, a boolean clinical `flag`, the cited `caveat`, and any quality flags. This contract is frozen as v0.1 and is the coupling point to the frontend and deployment layers.

5 Datasets

5.1 Healthy anchor: Spine-Generic

Spine-Generic [4] is an open, multi-vendor, multi-site collection of cervical T2 acquisitions from healthy adult volunteers. We use 12 controls spanning several vendors/sites (amu, beijingGE, brnoUhb, fslAchieva) as the healthy anchor. Because these are young healthy subjects, their canals are wide and their discs are well-hydrated — the correct reference for "normal."

5.2 Symptomatic cohort: MMCSD

MMCSD [19] (Yu et al., *Scientific Data*, `doi:10.1038/s41597-025-05403-z`; Synapse `syn63903115`) contains 250 surgical cervical-spondylosis patients with real NIfTI volumes (sagittal T1/T2, axial T2, axial CT) and, critically, *per-case labels*: cervical spondylotic myelopathy (CSM, $n = 108$), radiculopathy (CSR, $n = 122$), or mixed ($n = 20$), plus pre/post operative VAS neck-pain scores and *per-level lesion flags* (which segments C2–3 . . . C7–T1 carry a lesion). This per-case and per-level labelling is exactly the stratification information that other cervical MRI sets lack, which is why MMCSD is our symptomatic cohort. It is distributed under CC BY-NC-ND 4.0, so it is used for non-commercial validation only and neither the images nor any derived masks are redistributed. We validate on a balanced starter subset of 10 cases (5 CSM, 5 CSR) selected to carry mid-cervical lesions; the implications of this selection are discussed in Section 16.

5.3 Why MMCSD validates multiple groups, not just one

A common misconception is that a "spondylosis" dataset only tests spondylolisthesis. Spondyl*osis* is the broad degenerative umbrella (disc degeneration, osteophytes, canal stenosis, cord compression, loss of lordosis); spondyl*olisthesis* (vertebral slip) is a minor, incidental part of it. MMCSD's CSM cases are a canal/cord (Group 3) and cord-signal (Group 5.1) disease, and its CSR cases are a disc (Group 2) disease. The cohort therefore exercises Groups 2, 3, 4, and 5; it does *not* exercise the vertebral compression axis (Group 1 $\mathrm{H_a}/\mathrm{H_p}$), because spondylosis does not cause compression fractures.

5.4 Duke CSpineSeg and the ground-truth gap

Duke CSpineSeg [21] offers clinical cervical T2 with expert vertebra/disc masks but no per-case pathology labels and no per-case measurements; it is useful only for distribution-level comparison. Across repeated, exhaustive searches (multiple platform sweeps and multi-agent literature workflows) we confirmed that *no* public dataset pairs cervical MRI with per-case expert measurements — the persistent "KIND-B gap." The single dataset pairing cervical MRI with expert Cobb angles (F1000 "Data of Cervical Spine," $n = 77$ spondylosis) stores images as PACS screenshots, not segmentable volumes, so it serves only as a distribution reference. This absence is the reason our validation is threshold-crossing rather than per-case accuracy.

5.5 Data governance

NIfTI/DICOM volumes and derived masks are never committed to version control; they live in local storage and cloud drives outside the repository, and the `.gitignore` additionally blocks the cohort directories, mask folders, and clinical-label spreadsheets. Spine-Generic (open) may be shared as a viewer sample; MMCSD and Duke (CC BY-NC-ND) may not.

6 Methods: Segmentation

We delegate segmentation to three pre-trained engines, each producing masks for the structures it is strongest on, on a common per-case grid.

TotalSpineSeg (Groups 1, 2, 4). Run with `-iso`, TSS produces a step-2 multi-label mask with per-vertebra labels (canal = 2, C1= 11 . . . C7= 17, T1= 21. . .) and disc labels (C2–C3= 63 . . . C6–C7= 67, C7–T1= 71), plus a level map and an isotropic input volume. The verified label conventions (notably the C7–T1 disc = 71, and canal = 2 as used by our measurement code) were confirmed against real output.

Spinal Cord Toolbox (Group 3, Group 5.1). `sct_deepseg` segments the dural-sac/canal and the spinal cord as binary masks on the isotropic volume; `sct_process_segmentation -perslice -vert 2:7 -discfile <levels>` then yields per-slice AP diameters mapped to vertebral levels C2–C7. SCIseg (`sct_deepseg lesion_sci_t2`) provides the cord T2-lesion mask for Group 5.1. SCT runs are aligned to the TSS level map so canal and cord measurements share a level index.

SPINEPS (Group 4, C1 method). SPINEPS predicts vertebra instances *and* writes each vertebra's inferior-endplate sheet into the instance mask at label $100+X$ (C2= 102 . . . C7= 107; superior endplates at $200 + X$). These endplate voxels are the substrate for our best Cobb method (Section 10). SPINEPS requires `numpy==2.0.2`, incompatible with the TSS/nnU-Net stack, so it runs in a separate environment/session.

Engineering notes. Segmentation runs on GPU (Colab A100). A subtle but important failure mode we encountered: clearing `LD_LIBRARY_PATH` in a subprocess (a workaround for one tool's bundled libraries) silently disabled CUDA for TotalSpineSeg, forcing CPU inference and a $\sim$35$\times$ slowdown; the fix was to stop overriding the variable so PyTorch could find the CUDA driver. Such operational details are part of making a multi-tool pipeline actually run.

7 Methods: Group 1 — Vertebral Body

Group 1 measures, per vertebra C3–C7, the anterior/middle/posterior heights (H_a, H_m, H_p), the compression ratio $\mathrm{H_a/H_p}$, AP width, tilt, and inter-vertebral slip (spondylolisthesis). The accuracy of all height- and angle-derived outputs rests on correctly isolating the vertebral *body* and locating its endplates. Our method evolved through three corrections (detailed in Section 15).

Body isolation by canal-cut. On a mid-sagittal cervical slice the vertebral body and the posterior arch are a single fused blob, so connected-component and erosion heuristics leave a wedge-shaped body and a falsely low $\mathrm{H_a/H_p}$. We instead use a principled anatomical boundary: the body is everything *anterior to the spinal canal* (TSS canal label = 2). This canal-cut (`extract_body_via_canal`) removes the arch cleanly and is robust across subjects.

Heights by PCA + endplate-line fitting. Measuring height along the image axes fails on tilted, posteriorly tapered cervical bodies. We orient each body to its own principal axes (PCA in physical-mm space), then fit straight lines to the superior and inferior endplate boundaries and read the gap at the anterior/middle/posterior margins, with a thin-bin filter discarding rounded tips. This endplate-line formulation is tilt- and taper-invariant and is the same idea the literature finds superior to corner extrema [17, 3].

Compression screen. The clinical Genant grade [6] is retained as a standard, but the screening decision uses a data-driven cervical flag: H_a/H_p is compared to the measured healthy cohort norm 0.94 ± 0.13 (Spine-Generic, $n = 60$ vertebrae across 12 subjects, 3 vendors), flagging when $H_a/H_p <$ mean $-z \cdot$SD with $z = 2$ (≈ 0.68). The healthy norm is supported by triangulation against cervical anatomy literature [15, 8, 7] (posterior slightly > anterior, mean ≈ 0.90); we explicitly note that no like-for-like healthy cervical MRI H_a/H_p comparator exists, so this is plausibility, not gold-standard proof. The replaced in-code value (0.97 ± 0.02) was traced to an osteoporotic cohort and debunked (Section 15).

AP width, tilt, slip. AP width and tilt are treated as quality/sanity metrics (tilt over-flags on healthy at the in-code 20° cut, which we recommend recalibrating to ≈ 28°). Spondylolisthesis is measured from the posterior vertebral surface; it remains *experimental* because no supine-MRI presence threshold exists (the ≥ 2 mm cut is an upright-radiograph borrow [11]) and supine MRI under-measures functional slip.

Status. The endplate-line H_a/H_p screen is validated (Section 14); AP width/tilt are quality metrics; slip is experimental. In the production service tree, porting the endplate-line heights into `cervical_body_morphometry` (replacing the legacy corner method) is the one remaining Group 1 code task.

8 Methods: Group 2 — Intervertebral Disc

Group 2 measures, per disc C2–C3 through C7–T1: disc signal-intensity height, the disc-height index (DHI), posterior disc bulge, and a cervical signal-degeneration grade. The components form a chain (`disc_si_height` → `disc_height_index` → `disc_ap_bulge` → `pfirrmann_grade`), threading prior results.

Disc height and DHI. Disc height is measured at the midline band selected via the canal; DHI normalizes the disc middle height by the mean of the adjacent vertebral-body middle heights. The in-code absolute cut (DHI < 0.30) is uncited (an animal/lumbar borrow); the cervical-appropriate notion of reduced disc height is a >30% inter-level drop [14] or an absolute height < 3 mm, both requiring cross-level context.

Posterior bulge. Bulge is the posterior excursion of the disc beyond the vertebral margin. The correct reference is the tilted chord between adjacent posterior vertebral-body corners; a flat vertical back-wall mis-measures on lordotic necks. Cervical bulge thresholds derive from Nakashima et al. [13] (> 1 mm bulge); the often-cited 1.35 mm "cord-risk" figure is treated as unverified pending full-text confirmation.

Signal grade. Cervical disc degeneration is graded by the modified (Miyazaki) scheme [9], computed from a CSF-normalized nucleus signal ratio (requiring the raw T2 volume), *not* the lumbar Pfirrmann scheme. The ratio-to-grade cut-points have no published κ/AUC for cervical and are therefore scoped as a research-grade heuristic, surfaced for review.

Validation outcome and a detected bug. On the cohort, both DHI and bulge read backwards (healthy scored *worse* than the symptomatic cohort; Section 14). The DHI denominator is over-measured at the cervicothoracic junction (adjacent-VB middle heights of 20–27 mm versus the true ∼12–13 mm), collapsing DHI and over-firing the reduced-disc flag — the exact denominator discrepancy predicted by the component's author. Because Group 2 is a teammate-owned module tuned on a different (Duke) cohort, the bug is documented and flagged with its root cause and a candidate fix (robustify the adjacent-VB height at junction levels), rather than rewritten without the author and without ground truth. Group 2 is therefore *not validated* in this pass; the validation harness correctly identified it as broken.

9 Methods: Group 3 — Canal and Cord

Group 3 measures the functional (soft-tissue) canal / dural-sac AP minimum, the spinal cord AP diameter, the space available for the cord (SAC), and the Torg–Pavlov ratio. These are computed from the Spinal Cord Toolbox segmentations rather than from TSS, because SCT's cord and canal models are purpose-built and validated for soft-tissue cord/canal morphometry.

Computation. `sct_deepseg` produces binary canal and cord masks on the isotropic volume; `sct_process_segmentation -perslice -vert 2:7 -discfile <levels>` returns, per slice and per vertebral level, the AP diameter (and area, eccentricity, etc.). We take the per-level mean AP and the across-level minimum for the canal and cord; SAC is the per-level canal−cord difference, and we report its minimum. Because canal and cord are mapped to the same TSS level index, SAC is computed on matched levels.

Thresholds and the modality caveat. Normal cervical canal/SAC/Torg and vertebral-width percentiles come from healthy-population studies [1]; cord cross-section can be compared to the PAM50 normative database [16]. A key honesty point: several widely cited canal/SAC/Torg cut-points are *osseous canal / radiograph-derived* and over-flag when applied to MRI soft-tissue measurements. The Torg–Pavlov ratio (< 0.8) in particular fires in a large fraction of normal subjects on MRI and must never be used alone; the soft-tissue dural-sac floor is better placed near 9 mm to 10 mm; and SAC$<$ 3 mm is a radiograph-origin risk flag to be verified for MRI. The interpretation catalog encodes these caveats explicitly.

Status. Group 3 is the most strongly validated group: on real MRI, canal-AP minimum and SAC minimum separate healthy from symptomatic at $p = 0.0001$, and cord AP is significantly thinner in the symptomatic cohort (Section 14). Notably, on our healthy controls the SAC$<$ 3 and canal$<$ 10 cuts did *not* over-flag (0%), tempering — for this cohort — the over-flagging concern (though young controls have wide canals; see limitations).

10 Methods: Group 4 — Sagittal Alignment

Group 4 computes the global cervical Cobb angle (C2–C7 / C3–C7), per-level segmental angles, a posterior-tangent cross-check, and a lordosis classification. The Cobb angle is the most technically demanding measurement in the system, and its method is the project's main methodological story.

The corner problem. A Cobb angle is the angle between the top and bottom endplate tangents. Deriving those tangents from four/six corner landmarks is unstable on lordotic cervical necks because endplates are concave and sloped [3]: the legacy corner method read healthy necks as $-21°$ (kyphotic) when they are in fact lordotic. The literature is unambiguous that fitting a line to the full endplate boundary beats corner extrema [17].

Three methods, evaluated. We implemented and compared three Cobb methods on the 12 healthy necks (precision/coverage, since no GT exists):

1. **Canal-cut endplate-line** — fit the tangent to the canal-cut body's endplate boundary. Fixes the *sign* (now lordotic) but the C6/C7 endpoints remain noisy (the body is mis-shaped at the cervicothoracic junction).

2. **SPINEPS corpus border** — fit to the SPINEPS corpus (label 49) border. This was a *negative* result: noisier than canal-cut, because differencing two vertebrae amplifies a single bad per-vertebra fit.

3. **SPINEPS endplate voxels (C1)** — fit the line to SPINEPS' *own predicted endplate voxels* (instance label $100 + X$). This is the literature-validated object (the endplate itself, not the corpus border) and it won.

The C1 method. For each vertebra, the PCA major axis of the thin endplate sheet on the mid-sagittal slice is the endplate tangent; the Cobb angle is the angle between two vertebrae's inferior-endplate tangents, sign-calibrated to lordosis-positive. On the 12 healthy necks, C1 beats both alternatives on every span: C2–C7 15.4° ± 13.7° (matching the F1000 literature mean of 15.4°), and the C6–C7 cervicothoracic endpoint — the failure point of the other methods — drops to SD 5.9° (versus 18.5° for canal-cut). The benchmark ceiling for an automated cervical-T2 Cobb is Zhang et al. (MAE 2.44°) [20].

Reporting and a corrected assumption. Because absolute accuracy needs radiologist GT we lack, Group 4 reports alignment as a descriptive class (lordotic/straightened/kyphotic) for review, with the endpoint and supine caveats, rather than a hard-flagged magnitude. We also corrected a common false assumption: supine MRI does *not* read ∼5° less lordotic than standing for C2–C7 (the difference is not significant in the F1000 cohort), so a flat supine→standing offset should not be applied.

Status. C1 is the validated *method* (precision + correct lordosis): well-positioned controls read 15° to 20°, matching the standing-radiograph literature. As a *discriminator*, however, it is **not** validated: on a representative balanced cohort (26 healthy vs. 41 symptomatic) the healthy mean falls to 10.7°, essentially onto the symptomatic 8.0° ($d = 0.28$, AUC 0.57, $p = 0.32$; Section 14). The encouraging n=11 directional result ($d = 0.76$) was a lordosis-biased small sample. Cervical alignment is biologically too variable (healthy SD ∼10°) and supine positioning is a confound, so we report G4 as a validated *measurement*, not a screen.

11 Methods: Group 5 — Signal and Shape Screens

Group 5 screens for abnormal findings the geometric groups do not: vertebral-body compression/deformity and intramedullary cord signal change (myelomalacia). Both are screens that flag findings for physician review, never diagnoses.

5.2 Vertebral compression/deformity screen. This reuses the Group 1 H_a/H_p machinery (canal-cut body isolation + endplate-line heights) and flags H_a/H_p below the cohort-calibrated cut (≈ 0.68, mean −2SD). Scope honesty is essential: validated against RSNA-2022 cervical fractures, the geometric wedge metric has near-zero power on *non-compression* fractures (odontoid/facet/arch), so 5.2 is scoped explicitly as a vertebral-body *compression* screen, not a general fracture detector. Healthy false-positive rate fell from 17% to 0% after recalibration (Section 15).

5.1 Myelomalacia screen. Hand-rolled intensity thresholds cannot separate lesion from normal cord signal variation, so we adopt SCIseg [12] (`sct_deepseg lesion_sci_t2`) as the engine; sensitivity rests on its publication. Our contribution is the *specificity* check on healthy cords: 10/11 healthy cords were completely clean (the one false positive a small component at the cervicothoracic junction), ≈91% healthy specificity in our hands. Lesions are mapped to the cervical level they overlap and carried into the report.

5.3 / 5.4 scoped out. Tumour/mass detection (5.3) is scoped out — no public labelled cervical-tumour MRI exists to validate it. Post-surgical scar (5.4) is deferred — distinguishing scar from recurrent disc requires gadolinium-enhanced sequences not in the T2-only input. Both are listed under "not assessed" in the output contract.

The 5→6 contract. Group 5 emits a per-level findings document (compression screen + myelomalacia, each with value, status, and citable provenance, plus an explicit not-assessed list and no-diagnosis wording) that the interpretation layer consumes.

Status. Group 5 is validated: 5.2 (FP 17%→0%) and 5.1 (≈91% healthy specificity), with the compression scope and the no-MRI-comparator caveats documented.

12 Methods: Group 6 — Interpretation

The interpretation layer turns raw measurements into status, severity, and flags by comparing each value to a central, fully cited threshold catalog. It is deliberately separated from measurement extraction so that the science (the thresholds) lives in one auditable place.

The threshold catalog. `thresholds.py` holds one specification per measurement key (clinical name, unit, severity bands, flag threshold, citation, modality caveat, and a tag of raw/derived/quality). `classify(key, value)` walks the ordered bands and returns a standardized status (`within_reference` / `outside_reference` / `review_only` / `not_interpretable`), a per-measurement severity label, the boolean clinical flag (`true` iff the value lands in an "outside" band), and the locked citation and caveat. Where no cited threshold exists, the spec says so (citation `None`, status `review_only`) — the system never invents a number.

Provenance and honesty rules. Every clinical number carries a primary-source citation (PMID/DOI), copied verbatim from verified-research notes because automated literature search can hallucinate author names on real identifiers. Modality caveats are attached where a threshold is a radiograph/CT/lumbar borrow (e.g. Torg, the osseous canal cuts, the lumbar-origin disc rules), so the report never silently conflates a bony-canal threshold with a soft-tissue MRI measurement.

Quality versus clinical flags. A flag is classified as a quality/caution flag (not a patient abnormality) if its name matches markers such as `outlier`, `unreliable`, `approximate`, `low_confidence`, `misaligned`, `resolution`, or `warning`. This fixes an earlier mis-classification where a vertebral-tilt quality flag could be read as pathology.

Syndrome indicators. Provisional, advisory-only combination rules sketch myelopathy (canal narrowing + low SAC + cord signal at a level) and radiculopathy (weak on sagittal-only MRI, since foraminal dimensions are not measured) — each marked provisional and never diagnostic.

Threshold corrections folded in from validation research. A dedicated research pass produced corrections now reflected as caveats/bands: the dural-sac/SAC/Torg cuts are osseous/radiograph borrows that over-flag on MRI; the 1.35 mm disc "cord-risk" figure is unverified; the Miyazaki grade-IV is non-discriminating in isolation; and a prior MRI-vs-CT disc-height offset had been inverted.

Status. The engine and catalog are built and unit-tested; the end-to-end run is gated on the measurement validation that this paper reports, after which the gap-filling thresholds (MSCC/MCC, per-level segmental norms, etc.) are the remaining Group 6 work.

13 Validation Methodology

13.1 Why not sensitivity/specificity

Computing true per-case sensitivity/specificity requires a radiologist's per-case measurements as ground truth. No public dataset provides this for cervical MRI (Section 5). We therefore define "validated" operationally:

> *Healthy cases sit on the normal side of each cited threshold and never cross; a pathology-enriched cohort shifts/crosses into the abnormal side — with documented limits.*

This is a **threshold-crossing** + **distribution-separation** design. It demonstrates that each measurement reads the correct range, without claiming per-case diagnostic accuracy.

13.2 The load-bearing idea: disease-agnostic geometry

Each measurement reports a physical quantity (mm, degrees, ratio, present/absent) whose normal range is anchored on healthy-population literature, independent of disease. Three consequences make a single demonstration cohort sufficient for the groups it exercises:

1. Nothing is *trained* on the unhealthy cohort (segmenters are frozen, thresholds are cited) — there is no overfitting-to-spondylosis risk; the cohort is a demonstration set, not a training set.

2. A measurement crosses its threshold whenever the *anatomy* is abnormal, regardless of cause, so abnormal-detection generalizes across diseases producing the same anatomical change. We do not need a separate validation dataset per disease to trust that the canal-AP measurement trips on a narrow canal.

3. The healthy anchor is the real guarantee: if the normal range is calibrated, a value outside it is a flagged finding (for review, not diagnosis).

The corollary defines the genuine gaps: findings the system does not *measure* at all (e.g. a tumour mass that does not change canal/cord/disc/vertebra geometry; foraminal stenosis on sagittal-only MRI) would be missed; and an abnormal value is not by itself a symptomatic disease (specificity overlap), hence flag-for-review.

13.3 Cohorts, statistics, and figures

Healthy = 12 Spine-Generic controls; unhealthy = 10 MMCSD symptomatic cases (5 CSM, 5 CSR). For each case we compute a per-case summary metric per measurement (e.g. the across-level minimum canal AP), and compare the healthy and unhealthy distributions with a two-sided Mann–Whitney U test (a non-parametric test appropriate for small n and non-normal data). Per-measure strip plots (healthy vs. unhealthy, with the median and the p-value) are generated with matplotlib (open source). All figures and the numeric results are reproducible from `run_validation_master.py`.

13.4 What a correct result looks like per group

By construction we expect: strong separation where the disease lives (canal/cord stenosis → Group 3); weaker or absent separation where the metric is less specific (alignment → Group 4); a null result where the cohort lacks the pathology (vertebral compression → Group 1 on a non-compression cohort); and the harness should catch any metric that points the wrong way (a built-in bug detector). Section 14 shows each of these occurred — and, importantly, that the alignment lead which looked directional at small n *dissolved to a non-discriminator* once a representative, balanced cohort was used.

14 Results

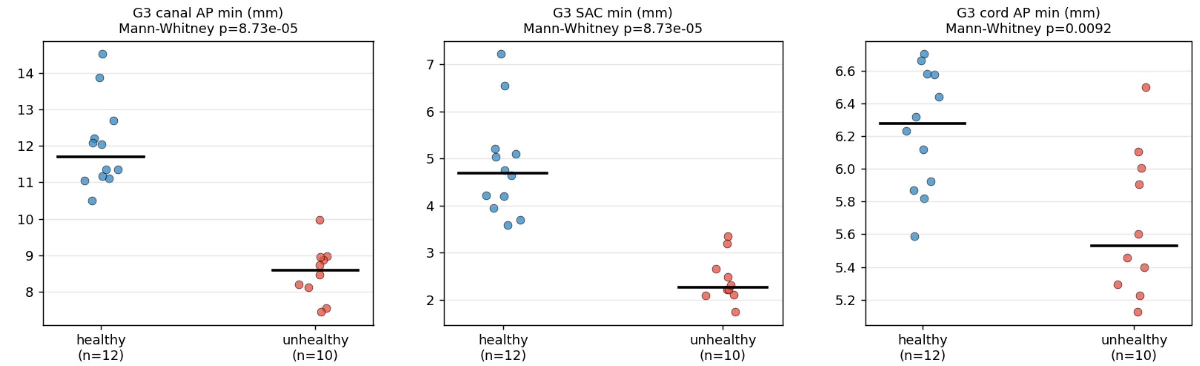
We report two stages. An *initial* threshold-crossing run on 12 healthy vs. 10 symptomatic necks (Section 14.1) established the headline canal/cord result and surfaced two leads (disc and alignment). We then scaled and refined those two leads (Section 14.4); the larger cohorts *revised* both conclusions, and we report the revised, honest verdicts. This progression — not stopping at the encouraging small-sample numbers — is itself a methodological result.

14.1 Initial run: canal/cord strong, compression null

Table 1 reports per-measure medians and the Mann–Whitney U p-value on 12 healthy vs. 10 symptomatic necks; Figure 2 shows the distributions.

Table 1: Initial threshold-crossing run: healthy (12) vs. symptomatic MMCSD (10). The physical-dimension (mm) metrics are scanner-immune and validate cross-dataset; the alignment and disc leads were subsequently re-evaluated on larger cohorts (Section 14.4).

Measure	**Healthy**	**Unhealthy**	p	**Verdict**
G3 canal AP min (mm)	11.7	8.6	0.0001	strong separation
G3 SAC min (mm)	4.7	2.3	0.0001	strong separation
G3 cord AP min (mm)	6.3	5.5	0.009	significant (thinner)
G1 H_a/H_p min	0.81	0.80	0.92	correctly null (by design)
G4 Cobb C1 C3–C7 (deg)	15.2	8.8	0.13	lead — re-evaluated below
G2 disc (DHI / bulge)	–	–	–	backwards (bug) — root-caused, fixed below



(a) Canal AP min (b) SAC min (c) Cord AP min

Figure 2: Initial-run healthy (blue) vs. unhealthy (red) distributions for the Group 3 physical-dimension metrics; black bar is the median, with the Mann–Whitney p-value. Canal AP and SAC separate cleanly.

14.2 Group 3 — canal/cord/SAC: strong separation (the headline)

Canal-AP minimum and SAC minimum both separate at $p = 0.0001$; the distributions barely touch (healthy canal floor 10.5 mm vs. unhealthy ceiling 9.97 mm). Cord AP is thinner in the symptomatic cohort ($p = 0.009$). On real MRI the canal/cord measurements read normal on healthy necks and cross into stenosis on a CSM/CSR cohort — the disease this cohort carries. Being physical-dimension (mm) metrics, they are scanner-immune and validate cross-dataset.

14.3 Group 1 — compression: correctly null

No difference (0.81 vs. 0.80, $p = 0.92$), zero compression flags in either cohort — the expected, correct outcome: spondylosis is not compression fracture, so vertebral heights are normal (true negatives). This empirically confirms the cohort does not exercise the compression axis (which needs a dedicated fracture dataset) and re-confirms the Group 5.2 specificity on all 12 healthy controls.

14.4 Scaled re-evaluation: the two leads, refined

The initial run left two leads — a directional alignment signal and a (buggy) disc signal. We took both to larger, design-appropriate cohorts. Crucially, the disc metrics are acquisition-sensitive intensity/ratio quantities, so they must be validated *within* one dataset, while alignment, though geometric, carries a positioning confound across datasets.

Group 4 — alignment: a validated *method*, not a discriminator. The initial directional result (healthy 15.2° vs. unhealthy 8.8°, $p = 0.13$, n=11) did *not* survive a representative sample. Adding multi-site healthy controls to reach **26 healthy vs. 41 symptomatic** dropped the healthy mean to 10.7° — essentially onto the symptomatic mean (8.0°) — with **Cohen $d = 0.28$, AUC 0.57,** $p = 0.32$: no separation. The original n=11 was a *lordosis-biased* draw (sites amu/beijingGE/fslAchieva, all 13° to 21°); the added sites are much flatter (nottwil 6°, balgrist 0.9°, one healthy control −12.7° kyphotic). Healthy cervical lordosis is simply too variable (SD ∼10°, range −13° to 32°), and supine *positioning* is a cross-cohort confound. The endplate-voxel *method* remains valid — well-positioned controls read 15° to 20°, matching the standing-radiograph literature, with precise endplate fits — but cervical alignment is a biologically weak, non-specific marker of myelopathy. We therefore report G4 as a validated *measurement*, not a screen. Reporting the encouraging n=11 result as "validated" would have been a small-sample artifact; the larger sample caught it.

Group 2 — disc: bugs fixed; geometric spread carries the only signal. The initial backwards reading was a real defect, since root-caused and corrected: vertebral heights were measured by corner extrema (reading H_a/H_p anterior-taller-than-posterior, $\sim$1.08, vs. the physiological $\sim$0.94) and the disc bulge used corner-derived reference corners; switching both to the validated endplate-line fit, and replacing a debunked absolute disc-height-index cut with a cited relative (>30% cross-level) rule, removed the artifacts (healthy bulge over-flag 60%→8%). We then ran the design-correct *within-MMCSD* test (49 symptomatic cases, 276 disc-levels, lesion vs. non-lesion, *level-stratified* to remove the confound that lesions cluster at wider mid-cervical levels). The disc *signal* grade does not discriminate (AUC 0.50, p = 0.93) and neither does posterior bulge (AUC 0.50; TotalSpineSeg does not segment protrusion) — two clean negatives. The discriminating quantity is disc *geometric spread*: the disc/vertebral-body AP-width ratio (AUC 0.62, p = 0.0018) and disc AP width (AUC 0.61 level-controlled; its raw AUC 0.79 was mostly the level confound). Group 2 is therefore *partially* validated — a modest but honest geometric discriminator, with signal and bulge documented as non-discriminators.

14.5 Overall

The validated-discriminator picture, after scaling, is clean and honest: **Group 3** (canal/SAC) separates strongly (p = 0.0001); **Group 2** disc/VB-AP-ratio separates modestly (AUC 0.62); **Group 4** alignment and the disc *signal* axis are non-discriminators; **Groups 1 and 5** are healthy-validated screens (with an under-tested compression arm). Four service-level measurement defects (vertebral-height corner method, an over-aggressive tilt cut, the disc bulge reference, and a debunked disc-height cut) were found and fixed during this work; the interpretation layer (Group 6) was wired end-to-end, including patient-demographic-aware thresholds. The recurring value is the methodology: the harness doubled as a bug detector, level-stratification and a representative sample caught two would-be false positives, and negative results were kept rather than buried.

15 Development Journey: Mistakes, Fixes, and Validation

The methodological core of this project is not a single algorithm but a disciplined sequence of corrections, each documented as *mistake → how it was found → the fix → how the fix was validated.* We summarize the sixteen entries below; the validation philosophy throughout is that, absent ground truth, "accurate" means "healthy lands in the published range and discriminates pathology, with documented limits," and that uncertain norms/methods are grounded in primary sources via adversarial literature search.

J1 — A phantom norm debunked. The in-code healthy H_a/H_p of 0.97 ± 0.02 was unsourced; a literature search traced it to an osteoporotic, all-female cohort. Measuring 12 healthy necks gave $0.94{\pm}0.13$, within the true healthy range [15, 8, 7]. Recalibration dropped the healthy false-positive rate from 17% to 0%.

J2 — Body isolation by canal-cut. Erosion/connected-component body isolation left a wedge and a stuck ratio; the fix was to define the body as everything anterior to the canal.

J3 — Endplate-line heights. Image-axis heights failed on tilted, tapered cervical bodies; PCA orientation + endplate-line fitting collapsed the spread and is robust to tilt and taper.

J4 — Honest scoping of the fracture screen. Validated against RSNA-2022, the wedge metric had near-zero power on non-compression fractures, so 5.2 was scoped to a vertebral-body compression screen, with the RSNA negative result banked.

J5 — Adopting SCIseg for myelomalacia. Hand-rolled intensity thresholds could not separate lesion from normal variation; we adopted the published SCIseg model [12] and reframed our job as the specificity check.

J6 — An over-claim, corrected, and the keystone. We initially mislabelled teammate measurements as "inaccurate" by treating quality flags as clinical ones; correcting that, and running the canonical code, localized the real problem to unstable corner-extrema landmarking (heights backwards, Cobb kyphotic, slip over-firing) — one keystone, not five bugs.

J7–J9 — The corner fix, grounded and implemented. An adversarial literature workflow confirmed endplate-line fitting beats corner extrema [17, 3]; implementing it fixed the Cobb sign (lordotic) and the heights, while the C6/C7 endpoints remained noisy — isolating body-isolation quality as the next lever.

J10 — Myelomalacia specificity measured. SCIseg on the healthy cords gave 10/11 clean (≈91% specificity), and the lesion-to-level mapping was wired end-to-end.

J11 — A negative result, kept. Swapping the body isolation for the SPINEPS corpus border made the Cobb *worse*; we recorded the negative rather than tuning to hide it, and noted that without GT we can only measure precision, not accuracy.

J12 — The endplate-voxel breakthrough. A research workflow found that SPINEPS already predicts the endplate itself (instance label $100 + X$); fitting the line to those voxels (method C1) beat both corner-cut and corpus on every span and matched the literature lordosis mean, closing the C6/C7 precision gap (SD 5.9°). It also corrected the false supine-offset assumption.

J13 — An integration fixture bug, caught by a test. Integrating Group 5 into the measurement service produced a failing test whose synthetic fixture had the anterior/posterior walls swapped for the RAS convention; fixing the fixture (not the science) restored a green suite — a reminder that synthetic fixtures must match real anatomy.

J14–J15 — Validation on real cohorts. The pipeline was run on 12 healthy and 10 symptomatic necks; Group 3 separated at $p = 0.0001$, Group 4-C1 directionally, and Group 1 was correctly null on a non-compression cohort (Section 14).

J16 — The harness as a bug detector. The same validation caught that Group 2's DHI and bulge read backwards and localized the cause to an over-measured denominator at the cervicothoracic junction — a real, sign-level bug a unit test on clean shapes would have missed.

J17–J21 — Owning the group code; G1 recalibrations. Consolidating ownership of the measurement modules, four service-level defects were measured and fixed on real cohorts: an over-aggressive vertebral-tilt cut (20° flagged 88% of healthy bodies; recalibrated to 45°), the vertebral-height corner method (reading $\mathrm{H_a}/\mathrm{H_p}$ backwards, $\sim$1.08; replaced by the endplate-line fit, $\sim$0.93), the disc bulge reference, and a debunked absolute disc-height cut (replaced by a cited relative rule). A resolution-robustness check confirmed the mm metrics are stable from 0.8 to 4 mm through-plane, while the canal-cut Cobb is not — a third argument for the endplate-voxel method.

J22–J23 — The disc, scaled and stratified. A 49-case within-MMCSD test (276 disc-levels, lesion vs. non-lesion) replaced the confounded cross-dataset comparison. The disc *signal* grade proved a non-discriminator (AUC 0.50), as did posterior bulge; the disc/VB AP-width ratio carried the only real signal (AUC 0.62). Critically, an apparent strong AP-width effect (AUC 0.79) collapsed to 0.61 under level stratification — the nuisance variable (lesions cluster at wider levels) had nearly produced a false headline.

J24–J26 — The alignment reversal. Group 4 Cobb was the sharpest lesson. At n=11 it showed a large directional effect (d=0.76); we resisted reporting it as validated and instead built a representative healthy cohort (26 multi-site controls). The effect *dissolved* (d=0.28, AUC 0.57, p=0.32): the original controls had been lordosis-biased, and supine positioning is a cross-cohort confound. Alignment is a non-specific marker; the larger sample, not a larger easy-side sample, settled it.

J25 — End-to-end and demographics. The disc group was wired into the orchestrator, patient age/sex were threaded through to the interpretation layer (with a cited sex-specific dural-sac cut), and the full pipeline was run end-to-end to produce structured reports cross-referenced against a deployed front-end render.

Process lessons. Several themes recur: (i) ground every clinical number in a primary source, because plausible-looking in-code constants can trace to the wrong cohort; (ii) separate measurement *method* quality from segmentation/body-isolation quality — they are different levers; (iii) keep negative results; (iv) without ground truth, distinguish precision from accuracy and never tune to make a method merely *look* tighter; and (v) a validation harness that compares healthy to pathology is also the cheapest, most honest bug detector available.

16 Limitations

No per-case ground truth. The central limitation: no public dataset pairs cervical MRI with per-case expert measurements, so we report distribution separation and precision, never per-case sensitivity/specificity or absolute accuracy (e.g. an absolute Cobb MAE). Closing this requires an institutional read (e.g. AUBMC), which we did not have.

Sample size and selection. The initial run used 12 healthy and 10 symptomatic necks, with the 10 unhealthy cases *selected to carry mid-cervical lesions*; part of the Group 3 sharpness is therefore by construction, and a random draw from the 250 MMCSD cases remains the necessary next test for G3. The disc and alignment leads were subsequently scaled (to 49 symptomatic cases for the within-MMCSD disc test, and to 26 healthy vs. 41 symptomatic for alignment), which is what *revised* those two conclusions; nonetheless all cohorts remain modest and single-institution on the symptomatic side.

Alignment is a non-specific marker, and the small-sample lesson. Group 4 Cobb illustrates a trap we explicitly avoided: an encouraging directional result at n=11 (d=0.76) dissolved to no separation (d=0.28, p=0.32) once the healthy arm was a representative multi-site sample. Healthy cervical lordosis is highly variable and supine positioning is a cross-cohort confound, so cervical alignment does not discriminate myelopathy in our data; we report it as a validated measurement, not a screen. The general limitation: with small samples and no ground truth, a "large effect" is a noisy estimate, and only a representative sample — not a larger one on the easy side — settles it.

Spectrum and cohort bias. MMCSD is a surgical (severe-end) cervical-spondylosis cohort, so claims are scoped to the cervical-spondylosis spectrum; generalization to mild disease or to non-degenerative pathology is not established. The healthy controls are young (wide canals), so the clean (0%) behaviour of the SAC<3/canal<10 cuts on healthy may not hold for an older clinical "normal."

Findings the system does not measure. Because measurements are geometric/signal detectors, a pathology that does not change the measured anatomy is missed — e.g. a tumour mass without geometric distortion, foraminal stenosis (not measurable on sagittal MRI), or inflammatory signal we do not screen for. These are scope limitations, stated explicitly; tumour (5.3) and scar (5.4) are out of scope.

The compression-fracture arm is under-tested. Spondylosis does not exercise the Group 1/5.2 compression axis, and no labelled cervical compression-fracture MRI dataset surfaced in repeated hunts; the existing 5.2 "unhealthy" evidence is degenerative (Duke DCM) and CT (RSNA, mostly non-compression). This is epidemiologically grounded, not a search artifact: the VerSe authors note that only thoracolumbar vertebrae were graded because fractures are rare and usually non-osteoporotic at the cervical spine (Löffler/Sekuboyina, PMC8082364). This axis remains the weakest and is a documented data gap; the one external lead is an access-blocked Penn cohort (Madi 2025, PMC11718528).

Absent measurement baselines (what we could not benchmark against). Several of the structures we measure have *no* published reference baseline at all, so even precision cannot be compared to a human standard: there is no cervical-MRI vertebral-body height (Ha/Hp) interobserver ICC, no cervical space-available-for-cord (SAC) millimetre ICC, no published cervical disc-AP/vertebral-body-AP ratio norm (our ≥ 1.10 cut is cohort-derived, with the mechanism — disc AP widening with degeneration — grounded in Machino 2021, PMID 34098133), and no healthy cervical-MRI Cobb ground-truth set (only an $n{=}77$ spondylosis MRI+Cobb reference exists). These are documented negatives from systematic searches, and each is a place where automation fills an unmeasured gap rather than beating a known number.

Threshold provenance and modality. Several thresholds are borrows from radiograph/CT/lumbar sources (Torg, some canal cuts, lumbar disc rules); they are flagged with caveats but their MRI soft-tissue calibration is not established. The disc-degeneration grade cut-points lack published κ/AUC for cervical and are research-grade. Spondylolisthesis has no supine-MRI presence threshold and is reported as experimental.

Group 2 partial; Group 6 now end-to-end. The disc-group defects were root-caused and fixed (endplate-line heights, endplate-corner bulge reference, a cited relative disc-height rule); the group is now *partially* validated — the disc/VB-AP-ratio separates modestly (AUC 0.62) while disc signal and posterior bulge are documented non-discriminators, and no per-case cutpoint is claimed. The interpretation layer (Group 6) was wired end-to-end and runs the full cited catalog with demographic-aware (sex-specific) thresholds; its per-case clinical accuracy still awaits an institutional read.

Slice thickness. Cervical T2 is typically 3 mm to 4 mm through-plane, so the 1 mm isotropic measurement grid is interpolated; the system surfaces a resolution-quality flag, but coarse acquisitions reduce confidence in fine measurements.

17 Deployment and Engineering

17.1 Service decomposition

The system is built as an external entry point (EEP) over internal processing services (IEPs): segmentation, measurements, interpretation, and reporting, each independently containerizable. The measurement service registers components in a dependency-ordered registry and instruments each with Prometheus metrics (`measurement_duration_seconds`, `measurement_results_total`, `measurement_pathology_flags_total`). The segmentation engines run on GPU; the measurement and interpretation layers are CPU-only, so the heavy GPU stage is isolated from the light, frequently changing measurement logic.

17.2 The data contract

A versioned data contract (v0.1) is the single coupling point between the science track and the frontend/infrastructure track: it freezes the shapes of the measurements dictionary, flags dictionary, per-component status, and the interpreted-measurement container (with the four-value status vocabulary), and proposes the case/job/upload/error/report envelopes. The frontend builds against a mock that matches these shapes, and the real measurement code drops in behind them unchanged. A companion contract specifies the NIfTI volume+mask delivery and label map for an interactive WebGL viewer.

17.3 Reproducibility and governance

Segmentation is reproducible from versioned Colab notebooks (TSS+SCT in one session, SPINEPS in a separate session because of the numpy pin). The measurement and validation steps are CPU-only and reproducible from committed scripts. Patient data (NIfTI/DICOM) and derived masks are never committed; licensed datasets (MMCSD, Duke; CC BY-NC-ND) are used for non-commercial validation only and not redistributed, while the openly licensed Spine-Generic sample may be shared (e.g. as a viewer fixture).

17.4 Development discipline

The project follows a strict session-handoff protocol (a living `SESSION_LOG` and a curated `DEVELOPMENT_JOURNEY`), granular commits, branch-per-feature with teammate-code changes routed through pull requests, and a hard no-secrets/no-patient-data-in-git rule. Every clinical claim carries a citation, and uncertain numbers are marked rather than invented — the same honesty discipline that the validation embodies.

18 Conclusion

MRI-ReportGenerator is an end-to-end, application-positioned pipeline that turns a single sagittal T2 cervical MRI into a structured, literature-grounded, no-diagnosis report. Its contribution is disciplined engineering rather than a new algorithm: composing three validated segmentation engines, organizing measurements into five clinically meaningful groups, grounding every threshold in a cited primary source with explicit modality caveats, and — most importantly — validating the assembled system honestly.

Because no public dataset pairs cervical MRI with per-case expert measurements, we adopted a threshold-crossing design built on the principle that measurements are disease-agnostic geometric/signal detectors anchored on healthy norms. On real cervical MRIs, the canal/cord group separated strongly ($p = 0.0001$) and the vertebral compression screen was correctly silent on a non-compression cohort. Two leads were then *revised* by larger, design-appropriate cohorts: the SPINEPS Cobb method reads correct lordosis but, on a representative multi-site healthy sample, does not discriminate myelopathy (a validated measurement, not a screen); and the disc group, after the harness detected and we fixed a sign-level bug, separates only modestly through disc geometric spread (AUC 0.62), with disc signal and bulge documented as non-discriminators. Four measurement defects were found and fixed and the interpretation layer was wired end-to-end — the value being not a clean scoreboard but a validation disciplined enough to overturn its own encouraging early numbers.

The honest limitations are clear: small, partly selected samples; a single (severe) disease spectrum; no per-case ground truth; an under-tested compression-fracture arm; a disc group only partially validated; and an interpretation layer whose per-case clinical accuracy still awaits an institutional read. The path forward is correspondingly concrete: scale Group 3 to a random MMCSD draw; pursue an institutional read (e.g. AUBMC) for true per-case accuracy and for the demographic-adjusted thresholds; secure a labelled cervical compression-fracture set for the one untested axis; and complete the deployed demonstration.

The broader lesson is methodological. In a domain without ground truth, the most valuable engineering is not a clever model but a transparent chain of evidence — cited thresholds, documented mistakes, kept negative results, precision distinguished from accuracy, and a validation that doubles as a bug detector. That chain, more than any single number, is what makes an automated clinical measurement tool trustworthy enough to put in front of a physician for review.

A Appendix A: Per-case validation data

All 22 cohort necks (12 healthy Spine-Generic, 10 MMCSD symptomatic). H_a/H_p min and flags from Group 1; Cobb from Group 4 (canal-cut and SPINEPS C1); canal/cord/SAC minima from Group 3; DHI and bulge medians from Group 2. Dashes are unmeasurable spans.

case	grp	H_a/H_p	flg	$Cobb_{cc}$	$Cobb_{C1}$	canal	cord	SAC	DHI	bulge
sub-amu01	H	0.80	0	nan	16.7	14.5	6.7	6.5	0.22	3.32
sub-amu02	H	0.81	0	56.2	21.2	13.9	6.7	7.2	0.17	5.26
sub-amu03	H	0.71	0	-3.2	8.3	11.1	5.9	4.6	0.22	0.62
sub-amu04	H	0.83	0	nan	15.2	11.2	5.8	4.7	0.22	3.03
sub-beijingGE01	H	0.77	0	nan	28.6	12.2	6.3	5.0	0.16	2.47
sub-beijingGE02	H	0.83	0	-7.2	12.1	11.3	6.6	3.6	0.21	0.62
sub-beijingGE03	H	0.89	0	34.2	22.1	11.4	6.4	3.7	0.25	2.27
sub-brnoUhb01	H	0.98	0	4.1	-0.0	12.7	6.6	5.1	0.24	4.00
sub-fslAchieva01	H	0.87	0	-15.2	32.1	10.5	5.9	3.9	0.30	5.12
sub-fslAchieva02	H	0.76	0	2.6	2.7	11.0	6.2	4.2	0.28	0.83
sub-fslAchieva03	H	0.80	0	-6.3	8.1	12.1	5.6	5.2	0.25	2.88
sub-fslAchieva04	H	0.78	0	-8.4	nan	12.0	6.1	4.2	0.24	3.58
mmcsd-csm-002	U	0.80	0	-3.2	4.3	10.0	6.1	3.4	0.28	0.35
mmcsd-csm-004	U	0.76	0	13.2	-14.9	7.6	5.4	2.1	0.35	1.12
mmcsd-csm-005	U	0.92	0	-12.1	8.3	8.5	5.6	2.2	0.26	1.69
mmcsd-csm-007	U	0.87	0	-21.3	9.3	8.2	5.3	2.1	0.25	0.28
mmcsd-csm-009	U	0.79	0	-21.8	15.6	9.0	6.5	1.7	0.31	0.30
mmcsd-csr-001	U	0.91	0	-14.8	-4.2	7.4	5.2	2.2	0.32	0.05
mmcsd-csr-003	U	0.74	0	-10.9	0.5	8.1	5.5	2.7	0.22	0.81
mmcsd-csr-006	U	0.81	0	-17.0	10.1	8.9	6.0	2.3	0.23	1.20
mmcsd-csr-008	U	0.83	0	-13.2	13.7	9.0	5.9	2.5	0.25	1.01
mmcsd-csr-012	U	0.71	0	nan	17.4	8.7	5.1	3.2	0.21	2.18

Columns: H_a/H_p min (Group 1 compression ratio minimum across C3–C7), flg (compression-screen flags), Cobb_{cc} (canal-cut Cobb C3–C7, deg), Cobb_{C1} (SPINEPS endplate Cobb C3–C7, deg), canal/cord/SAC (Group 3 across-level minima, mm), DHI and bulge (Group 2 medians).

B Appendix B: The cited-threshold catalog

Every measurement key interpreted by Group 6, with its severity bands, unit, tag, and the locked primary-source citation. Reproduced from `services/interpretation/thresholds.py`. Identifiers are copied verbatim from verified-research notes.

key (unit, tag)	clinical name	bands	citation
`vb_hahp_ratio` (ratio, derived)	vertebral-body anterior/posterior height ratio (Ha/Hp) compression screen	normal [0.81,+inf); borderline [0.68,0.81); compression_screen_positive [-inf,0.68)	Tan 2004 (Eur Spine J, PMC3476578); Lee 2012 (PMC3393857); Kaur 2025 (J Human Anat); Chen 2013 (PLoS One, PMC3859485); Nell 2019 (PMC6764695)
`spondy_slip_mm` (mm, raw)	vertebral slip magnitude (anterolisthesis / retrolisthesis)	neutral [-inf,1); borderline [1,2); slip_present_screen [2,+inf)	Murakami 2020 (PMID 32591548); Murata 2019 (PMID 30899028); de Dios 2023 (cervical-MRI slip, MDC 1.5 mm); White 1975 (PMID 1132209, 3.5 mm = instability)
`posterior_bulge_mm` (mm, raw)	posterior disc bulge excursion	no_bulge [-inf,1); bulge_present_nonspecific [1,+inf) (review)	Nakashima 2015 (PMID 25584950, n=1211; >1 mm = bulge but NON-SPECIFIC, 87.6% of asymptomatic → review). The 1.35 mm cord-risk band was REMOVED (unverified / likely confabulated, §A.6). Documented NEGATIVE: posterior bulge does not discriminate (AUC 0.50, J23) – TSS segments to anatomical borders, not protrusion
`disc_vb_ap_ratio` (ratio, derived)	disc AP width / adjacent vertebral-body AP width ratio	within_cohort [-inf,1.10); disc_spread_review [1.10,+inf)	COHORT-DERIVED cut (NO published cervical disc/VB-AP-ratio norm exists – search NEGATIVE); mechanism literature-grounded: disc AP diameter widens with degeneration (Machino 2021, PMID 34098133, n=1211 MRI); Fardon v2.0 2014 (Spine J 14(11):2525-2545, doi:10.1016/j.spinee.2014.04.022, lumbar bulge-margin borrow). G2's discriminator (AUC 0.62, level-controlled, J23); review-only, no per-case GT

key (unit, tag)	clinical name	bands	citation
`pfirrmann_grade` (grade, derived)	cervical disc signal grade (Miyazaki, modified Pfirrmann)	grade_I [1,2); grade_II [2,3); grade_III [3,4); grade_IV [4,5); grade_V [5,6)	Miyazaki 2008 (PMID 18525490) cervical modified grading (NOT lumbar Pfirrmann); CSF-normalized ratio cervical-validated (Liu 2023 PMID 37156851, Watanabe 2025 PMID 39645168). DOI / Grade I-V table PDF check pending (human-gated)
`DHI` (ratio, derived)	disc height index	review-only (no cited cut)	DHI<0.30 DEBUNKED (uncited, animal-lumbar borrow). Reduced disc height = >30% drop vs adjacent (Suzuki 2018) or absolute <3 mm (van Santbrink 2026); closest cervical DHI formula = Machino 2021 (PMID 34098133, paywalled). No validated absolute cervical DHI cut-point – pending Phase-4.
`dural_sac_AP_min` (mm, raw)	functional canal / dural-sac AP minimum (SCT)	normal [13,+inf); borderline [10,13); stenosis_provisional [-inf,10)	Nell 2019 (PMC6764695, C2-C7 healthy age/sex percentiles); stenosis bands >13/10-13/<10 mm are a clinical convention (Ulbrich 2014; Thelen 2019/SHIP) of OSSEOUS-canal / radiograph origin – pending Phase-4 MRI soft-tissue confirmation
`cord_AP` (mm, raw)	spinal cord AP diameter	review-only (no cited cut)	Valosek 2024 PAM50 healthy-control database via SCT -normalize-hc (age/sex). No fixed mm cut hard-coded here
`SAC` (mm, derived)	space available for the cord (canal AP - cord AP)	normal [3,+inf); high_risk [-inf,3)	SAC <3 mm = high compression risk (plan-cited Fehlings 2015 / Nouri 2016) – radiograph-origin, verify for MRI (pending Phase-4); Nell 2019 (PMC6764695) healthy percentiles
`Torg_Pavlov_ratio` (ratio, derived)	Torg-Pavlov ratio (canal AP / vertebral-body AP)	normal [0.8,+inf); developmental_stenosis_screen [-inf,0.8)	Torg 1987 / Pavlov 1987: ratio <0.8 = developmental canal stenosis – RADIOGRAPH origin; MRI thresholds may need adjustment (pending Phase-4)
`Cobb_C3_C7` (deg, derived)	global cervical Cobb angle (C3-C7, lordosis-positive)	lordotic [10,+inf); straightened [0,10); kyphotic [-inf,0)	NASSJ 2025 (PMC12744292, asymptomatic C2-C7 ~18.7 deg, range ~10-35 deg); plan also cites Yukawa 2018 (~13.9 +/- 12.3 deg). Lordosis-positive.
`segmental_angle` (deg, raw)	per-level segmental angle	review-only (no cited cut)	–
`posterior_tangent_C3_C7` (deg, derived)	posterior tangent angle (C3-C7)	review-only (no cited cut)	–

key (unit, tag)	clinical name	bands	citation
`AP_width` (mm, quality)	vertebral-body AP width	within_sanity [12,22); ap_width_outlier [-inf,12); ap_width_outlier [22,+inf)	Nell 2019 (PMC6764695) healthy per-level AP-width percentiles for clinical comparison; the in-code 12-22 mm is an ENGINEERING sanity window, not a clinical pathology threshold
`tilt_deg` (deg, quality)	vertebral-body tilt (vs global SI axis)	review-only (no cited cut)	–
`myelomalacia` (present, derived)	cord T2-hyperintensity screen (myelomalacia)	none [-inf,0.5); signal_anomaly_present [0.5,+inf)	SCIseg (sct_deepseg lesion_sci_t2); Naga Karthik 2024 (PMC11065035). Measured healthy specificity 10/11 (~91%)

C Appendix C: Additional figures

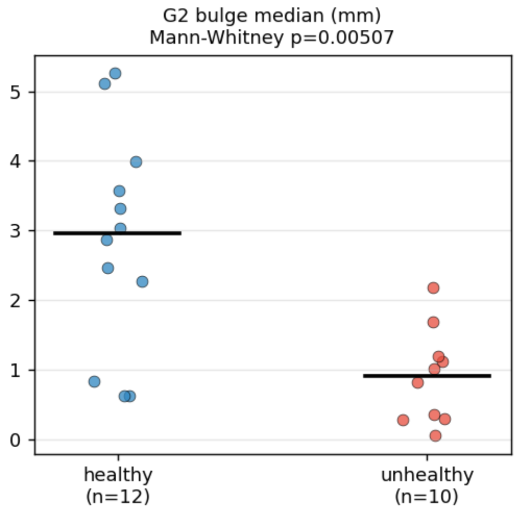


Figure 3: Group 2 posterior disc bulge (median per case), healthy vs. unhealthy. The metric reads *backwards* — healthy shows a higher bulge than the symptomatic cohort — one of the two sign-level disc-metric bugs the validation harness detected (Section 14, Section 8). Retained as evidence rather than hidden.

The remaining per-measure strip plots appear in Figure 2 (Section 14). All figures are regenerated from `run_validation_master.py` (matplotlib), and the architecture diagram (Figure 1) is drawn in TikZ so it requires no external image file.

D Appendix D: The data contract (v0.1)

The single coupling point between the measurement/science layer and the frontend/deployment layer. The measurement/interpretation core is frozen; the case/job/upload/report envelopes are proposed.

D.1 Measurements and flags

`measurements` is `{measurement_key: {level_or_pair: number}}`; `flags` is `{flag_key: {level_or_pair: bool}}`. Levels are `C3`. . . `C7`; disc/segmental keys use pairs (e.g. `C3-C4`). Representative keys by group:

key	unit	keyed by	group
`AP_width`	mm	level	G1
`H_anterior / H_middle / H_posterior`	mm	level	G1
`vb_hahp_ratio`	ratio	level	G1/G5
`tilt_deg`	deg	level	G1 (quality)
`spondy_slip_mm`	mm	pair	G1
`posterior_bulge_mm`	mm	pair	G2
`disc_height_index (DHI)`	ratio	pair	G2
`pfirrmann_grade`	grade	pair	G2
`dural_sac_AP_min`	mm	level	G3
`cord_AP`	mm	level	G3
`SAC`	mm	level	G3
`Cobb_C3_C7`	deg	span	G4
`segmental_angle`	deg	pair	G4
`myelomalacia`	present	level	G5

D.2 Interpreted measurement container

Each (measurement × level) is wrapped as `{measurement, level, value, unit, status, severity, flag, demographics_used, quality_flags, caveat}`. `status` is one of the four standardized values `within_reference`, `outside_reference`, `review_only`, `not_interpretable`; `flag` is true iff `status` is `outside_reference`; `severity` is a per-measurement label; `caveat` is the cited modality caveat.

D.3 Quality versus clinical flags

A flag is quality/caution (not pathology) iff its name contains any of `outlier`, `unreliable`, `approximate`, `low_confidence`, `misaligned`, `resolution`, or `warning`.

D.4 Segmentation viewer

The viewer overlays a NIfTI volume and the TotalSpineSeg multi-label mask on a shared grid. Label map (cervical subset): canal/cord at labels 1/2 (note the verified convention: label 2 is the canal in the measurement code); vertebrae C1–C7 = 11–17, T1 = 21; discs C2–C3 = 63 . . . C6–C7 = 67, C7–T1 = 71. The dedicated SCT binary cord/canal masks are preferred for the canal/cord overlay.

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